

WATER

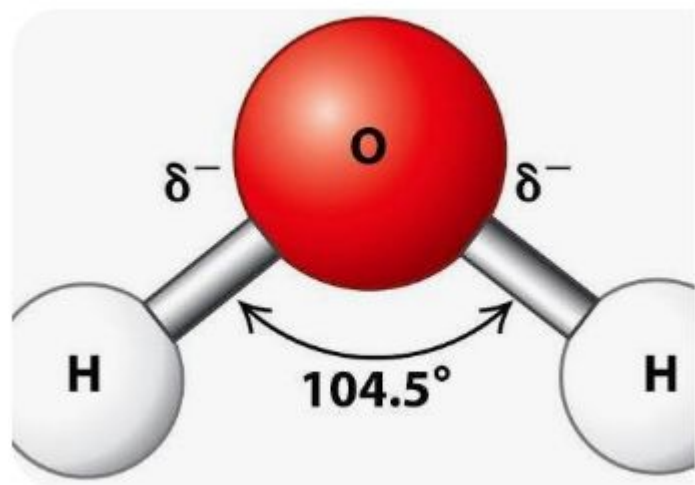
Water is the most abundant inorganic compound of the living cells.

It constitutes 60-95% of a living system and 80-95% of the protoplasm.

Water is found in two forms in a cell i.e. free water and bound water.

Water available for metabolic process is free water whereas water loosely attached to protein molecules by hydrogen bonds is bound water.

Structure of water



The structure of a water molecule (H_2O) consists of one oxygen atom covalently bonded to two hydrogen atoms, forming a bent or V-shape. This V-shape, along with the oxygen atom's higher electronegativity, makes water a polar molecule with a partial negative charge on the oxygen and partial positive charges on the hydrogens. These partial charges cause water molecules to attract each other through hydrogen bonds, giving water its unique properties. 

Functions(biological roles) of water

Water shows following biological functions:

1. Solvents: water acts as a solvent for many inorganic and organic and organic compounds. So, it is commonly called universal solvent.
2. Dispersion medium : Water serves as a good dispersion medium for the colloids of cells.
3. Ionizer: Water help in ionization of inorganic compounds present in the cells.
4. Due to low density water permit rapid diffusion of molecules in cytoplasm.
5. Transport medium: Water transport food and nutrients from one part to the other part of organisms.
6. Metabolites: Water help in cellular reactions occur in solution.
7. Turgidity: Water help in maintenance of turgidity in growing cells.
8. Movement: The gain and loss of water from cells is responsible for movement of plants parts.
9. Raw material: Water is raw material for photosynthesis.
10. Germination: Water is needed for germination of seeds and spores.

LIPIDS or FATS

Lipids are a heterogeneous group of fats and fatlike substances. They are insoluble in water and soluble in non-polar solvents like ether, acetone, chloroform etc.

They are made up of carbon, hydrogen, and oxygen . Oxygen content is very low.

Some lipids also contain small amount of phosphorus , nitrogen and Sulphur.

Fatty acids

Fatty acids are organic acids having hydrocarbon chains of 4-30 carbon atoms that ends in a carboxylic group.

Plants can form all types of fatty acids.

Animals can form all fatty acids except linoleic and arachidonic acids.

The fatty acids which are not synthesized in animals are called essential fatty acids which must include in diet . They are present in oils of sunflower, groundnut etc.

Types of fatty acids

Fatty acids are of two types saturated and unsaturated fatty acids.

Differences between saturated and unsaturated fatty acids

Feature	Saturated Fatty Acids	Unsaturated Fatty Acids
Bonds	Only single bonds between carbon atoms	One or more double bonds between carbon atoms
Molecular Shape	Straight, linear, and unbranched	Bent or kinked due to double bonds
State at Room Temp.	Solid	Liquid
Melting Point	Higher	Lower
Primary Sources	Animal fats (butter, meat) and some oils (coconut)	Plant-based oils (olive, canola), nuts, seeds, and fish oil
Health Impact	Can increase LDL cholesterol; linked to heart disease	Can lower LDL cholesterol; considered heart-healthy

Types of lipids

Lipids are of three types :Simple , compound and derived lipids.

1.Simple lipids:

They are the esters formed by the combination of fatty acids and alcohol without any additional groups.

They are neutral fats, oils or plant fats, wax , cutin and suberin.

2. Compound lipids: They are lipids in which lipids join with non lipids compounds. Like glycolipids ,lipoproteins ,phospholipids .

3. Derived lipids: They are lipid like chemicals steroids/sterols(cholesterol).

Function(biological significance) of lipids

Lipids have the following biological significances:

1. Lipids are the main source of energy for living beings.
2. Lipids are storage food of both plants and animals.
3. Phospholipids, glycolipids and lipoprotein are the structural components of cell membrane. Cholesterol is the component of cell membrane of animals. Lipoprotein are structural component of blood, milk and egg.
4. Unsaturated oils present in spores and seeds protect the embryo from ultraviolet radiations during their dispersal through air.
5. Subcutaneous fats present below skin helps in heat insulation.
6. Vital organs of the body like eyeballs, kidney, gonads are surrounded by shock absorbing cushions of fats.
7. Fats are solvent of vitamins A, D, E and K .
8. Cuticle made up of cutin and wax reduces the rate of cuticular transpiration.
9. Cholestrol produes sex hormones like progestrone, estrogen and testosterone.

MINERALS

Minerals are inorganic substances that are present in small amounts and regulate various metabolic functions of organisms.

They are present only about 2-3% of the cell but they are essential in different metabolism, growth and development of body.

Categories of minerals

Plants contain about 40 different elements.

Some of them are necessary for normal growth and development of plants, which are called essential elements. They are 17 in number.

Rest of the elements are called as nonessential elements, which are not necessary for growth and development.

Essential elements may be classified into two groups based on their concentration required.

They are Macronutrients and Micronutrients.

Macronutrients

- **What they are:** The main nutrients that provide energy and are needed in larger quantities, measured in grams. [🔗](#)
- **Examples:**
 - **Carbohydrates:** The body's primary energy source. [🔗](#)
 - **Proteins:** Essential for building and repairing tissues, and for producing enzymes and hormones. [🔗](#)
 - **Fats:** Provide a concentrated source of energy, protect organs, and help absorb certain vitamins. [🔗](#)

Micronutrients

- **What they are:** Vitamins and minerals that are needed in much smaller amounts, measured in milligrams or micrograms. [🔗](#)
- **Examples:**
 - **Vitamins:** Organic compounds that are crucial for many bodily functions, such as vitamin A, C, D, and B vitamins. [🔗](#)
 - **Minerals:** Inorganic substances like calcium, iron, zinc, and potassium that are vital for health. [🔗](#)

Differences between macro and micro molecules

- **Importance:** They support metabolism, the immune system, brain function, and countless other processes. 
- **Source:** You get micronutrients from the same foods that provide macronutrients. A diverse diet is key to obtaining a wide range of these essential vitamins and minerals. 

Feature	Micromolecules	Macromolecules
Size	Small	Very large
Molecular Weight	Low	High (often >10,000 Daltons)
Structure	Simple molecules	Polymers composed of repeating monomer subunits
Examples	Amino acids, glucose, water, ATP, nucleotides	Proteins, nucleic acids (DNA, RNA), carbohydrates (polysaccharides)
Function	Building blocks, energy carriers	Primary structural and functional components of cells

Specific roles and deficiency symptoms of essential mineral elements

Mineral	Role in Plants	Deficiency Symptoms
Nitrogen (N)	Essential for chlorophyll, amino acids, and proteins; promotes leafy growth and photosynthesis	Uniform yellowing of older leaves (chlorosis), stunted growth, small leaves, and weak roots
Phosphorus (P)	Crucial for energy transfer (ATP), root development, and flowering	Stunted growth, delayed maturity, and a dark green or purplish-red discoloration on leaves and stems
Potassium (K)	Regulates water balance, enzyme activation, and photosynthesis	Yellowing and scorching (necrosis) starting at the leaf margins, with symptoms progressing to the upper leaves
Calcium (Ca)	Important for cell wall structure, membrane function, and signaling	Distorted or stunted new growth; affected leaves can be stunted and have irregular shapes
Magnesium (Mg)	The central atom in the chlorophyll molecule	Yellowing between the veins (interveinal chlorosis) on older leaves, leaf curling, and impaired photosynthesis
Iron (Fe)	Essential for chlorophyll synthesis and electron transport	Yellowing between the veins (interveinal chlorosis) on younger leaves, which can

Zinc (Zn)	Involved in enzyme function and growth hormone production	Smaller leaves, interveinal chlorosis, and a "rosetting" effect (leaves are clustered tightly at the stem tip)
Manganese (Mn)	Activates enzymes and is involved in photosynthesis and respiration	Interveinal chlorosis in younger leaves, which may develop white or gray spots, and the leaves may become cupped or twisted
Boron (B)	Plays a role in cell wall structure, cell division, and sugar transport	Thickened, brittle, and distorted new leaves; abnormal flower and fruit development
